

Architecture Selection & Justification

Level-3

SRN:PES1UG24CS111

NAME : BHADRA R S

System: Railway Reservation System – Train Scheduling, Seat Allocation and Ticket Management

Architecture Selection

We chose Layered Architecture for the Railway Reservation System.

Architectural Choice — Two Reasons

- 1. Clear separation of responsibilities:* The system has distinct functionality for user interaction, reservation/booking logic, train and schedule management, and persistent data. A layered architecture separates these responsibilities into identifiable modules, making the system easier to understand and maintain.
- 2. Suitable for a standalone C/C++ application:* The requirements specify a standalone application using C/C++ with local files or an embedded database and no external network interface. Layered architecture fits this structure without introducing the operational and communication complexity of a distributed architecture.

Security Advantage

Layer separation supports role-based access by keeping passenger, reservation-staff and administrator operations behind controlled business-logic interfaces. This helps prevent users from directly accessing administrative or reservation-management functionality, supporting the stated role-based security requirement.

Performance Benefit

Because the initial system is standalone and uses local files or an embedded database, the layered design avoids unnecessary network communication between services. Keeping search, booking, seat allocation and data access within the application can provide a direct execution path and help the system meet its requirement that 95% of train-search queries return within 2 seconds under the defined

Railway Reservation System - Layered Architecture

